

International Economic Relations

ECON 308

Specific Factors and Income Distribution
(Chapter 4 Part 1)

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Learning Objectives

- 4.1** Understand how a mobile factor will respond to price changes by moving across sectors.
- 4.2** Explain why trade will generate both winners and losers in the short run.
- 4.3** Understand the meaning of gains from trade when there are losers.
- 4.4** Discuss the reasons why trade is a politically contentious issue.
- 4.5** Explain the arguments in favor of free trade despite the existence of losers.

Introduction

- If trade is so good for the economy, why is there such opposition?
- Two main reasons why international trade has strong effects on the distribution of income within a country:
 - Resources cannot move immediately or costlessly from one industry to another.
 - Industries differ in the factors of production they demand.

The Specific Factors Model (1 of 4)

- The **specific factors model** allows trade to affect income distribution.
- Assumptions of the model:
 - Two goods, cloth and food.
 - Three factors of production: labor (L), capital (K), and land (T for terrain).
 - Perfect competition prevails in all markets.

The Specific Factors Model (2 of 4)

- Cloth produced using capital and labor (but not land).
- Food produced using land and labor (but not capital).
- Labor is a mobile factor that can move between sectors.
- Land and capital are both specific factors used only in the production of one good.

The Specific Factors Model (3 of 4)

- How much of each good does the economy produce?
- The production function for cloth gives the quantity of cloth that can be produced given any input of capital and labor:

$$Q_c = Q_c(K, L_c)$$

- Q_c is the output of cloth
- K is the capital stock
- L_c is the labor force employed in cloth

The Specific Factors Model (4 of 4)

- The production function for food gives the quantity of food that can be produced given any input of land and labor:

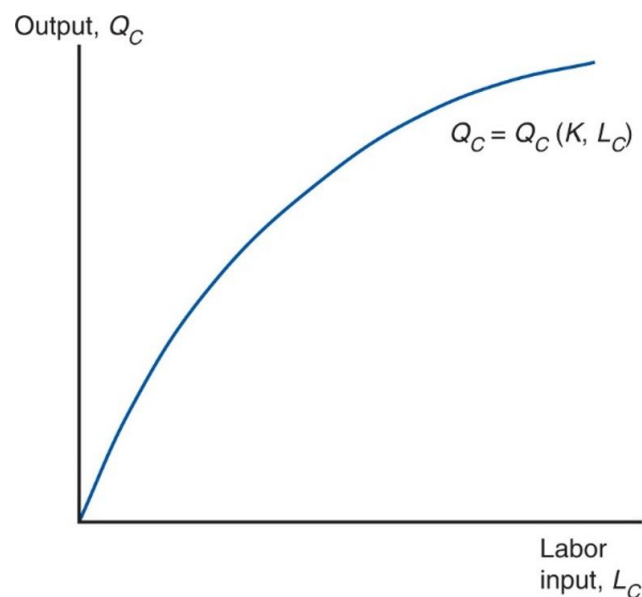
$$Q_F = Q_F(T, L_F)$$

- Q_F is the output of food
- T is the supply of land
- L_F is the labor force employed in food

Production Possibilities (1 of 6)

- How does the economy's mix of output change as labor is shifted from one sector to the other?
- When labor moves from food to cloth, food production falls while output of cloth rises.

Figure 4.1 The Production Function for Cloth

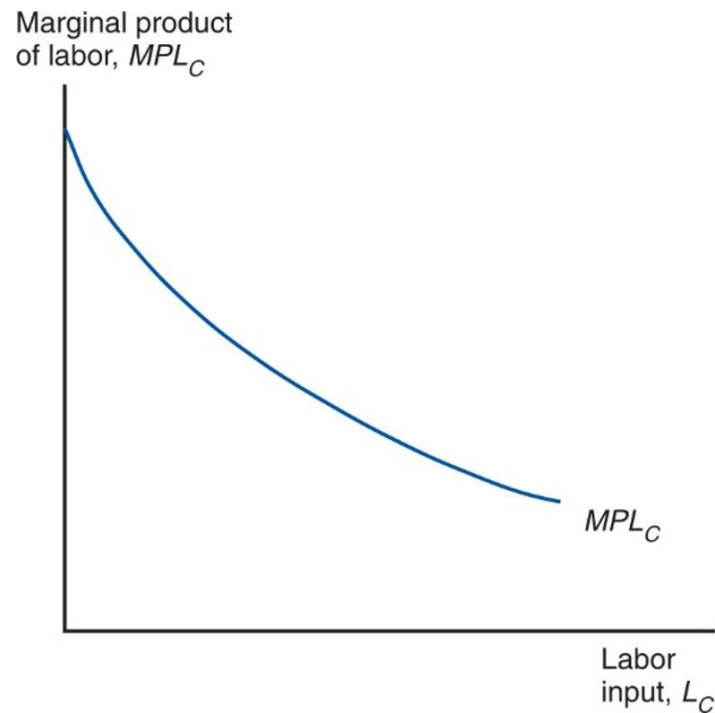


The more labor employed in the production of cloth, the larger the output. As a result of diminishing returns, however, each successive person-hour increases output by less than the previous one; this is shown by the fact that the curve relating labor input to output gets flatter at higher levels of employment.

Production Possibilities (2 of 6)

- The shape of the production function reflects the law of **diminishing marginal returns**.
 - Adding one worker to the production process (without increasing the amount of capital) means that each worker has less capital to work with.
 - Therefore, each additional unit of labor adds less output than the last.
- Figure 4.2 shows the **marginal product of labor**, which is the increase in output that corresponds to an extra unit of labor.

Figure 4.2 The Marginal Product of Labor



The marginal product of labor in the cloth sector, equal to the slope of the production function shown in Figure 4.1, is lower the more labor the sector employs.

Production Possibilities (3 of 6)

- For the economy as a whole, the total labor employed in cloth and food must equal the total labor supply:

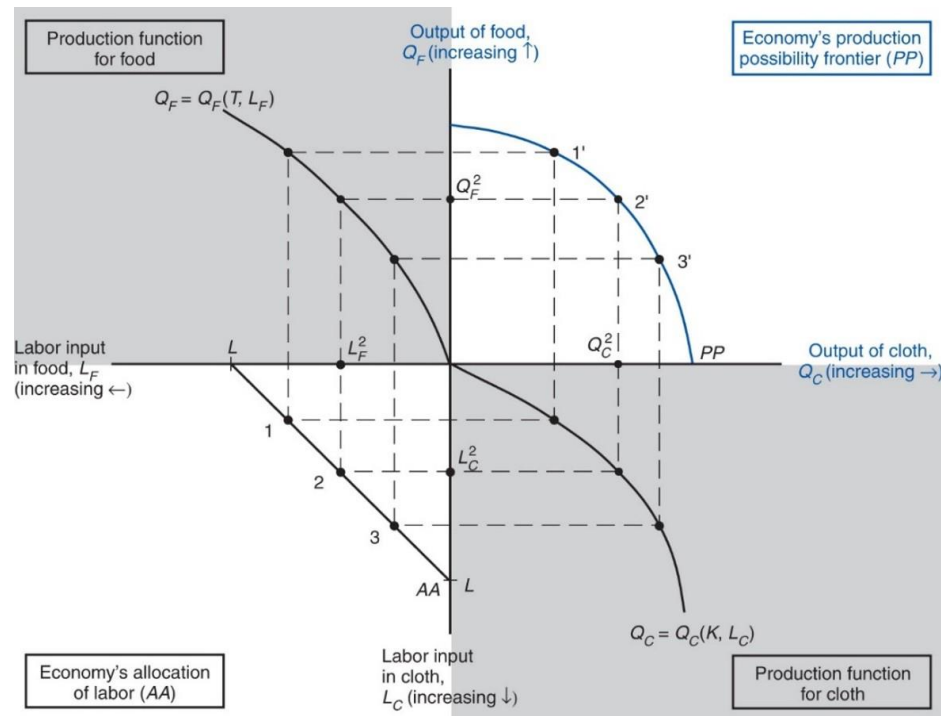
$$L_C + L_F = L$$

- Use these equations to derive the **production possibilities frontier** of the economy.

Production Possibilities (4 of 6)

- Use a four-quadrant diagram to construct production possibilities frontier in Figure 4.3.
 - Lower left quadrant indicates the allocation of labor.
 - Lower right quadrant shows the production function for cloth from Figure 4.1.
 - Upper left quadrant shows the corresponding production function for food.
 - Upper right quadrant indicates the combinations of cloth and food that can be produced.

Figure 4.3 The Production Possibility Frontier in the Specific Factors Model



PP in the upper-right quadrant shows the economy's production possibilities for given supplies of land, labor, and capital. Due to diminishing returns, PP is a bowed-out curve instead of a straight line.

Production Possibilities (5 of 6)

- Why is the production possibilities frontier curved?
 - Diminishing returns to labor in each sector cause the opportunity cost to rise when an economy produces more of a good.
 - Opportunity cost of cloth in terms of food is the slope of the production possibilities frontier—the slope becomes steeper as an economy produces more cloth.

Production Possibilities (6 of 6)

- Opportunity cost of producing one more yard of cloth is $\frac{MPL_F}{MPL_C}$ pounds of food.
 - To produce one more yard of cloth, you need $\frac{1}{MPL_C}$ hours of labor.
 - To free up one hour of labor, you must reduce output of food by MPL_F pounds.
 - To produce less food and more cloth, employ less in food and more in cloth.
 - The marginal product of labor in food rises and the marginal product of labor in cloth falls, so $\frac{MPL_F}{MPL_C}$ rises.

Prices, Wages, and Labor Allocation (1 of 10)

- How much labor is employed in each sector?
 - Need to look at supply and demand in the labor market.
- Demand for labor:
 - In each sector, employers will maximize profits by demanding labor up to the point where the value produced by an additional hour equals the marginal cost of employing a worker for that hour.

Prices, Wages, and Labor Allocation (2 of 10)

- The demand curve for labor in the cloth sector:

$$MPL_C \times P_C = W$$

- The wage equals the value of the marginal product of labor in manufacturing.

- The demand curve for labor in the food sector:

$$MPL_F \times P_F = W$$

- The wage equals the value of the marginal product of labor in food.

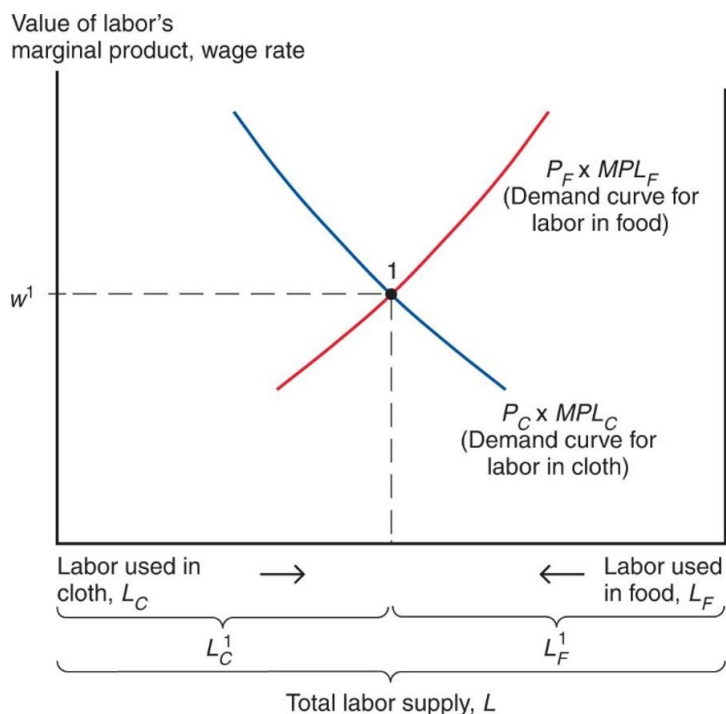
Prices, Wages, and Labor Allocation (3 of 10)

- Figure 4.4 represents labor demand in the two sectors.
- The demand for labor in the cloth sector is MPL_C from Figure 4.2 multiplied by P_C
- The demand for labor in the food sector is measured from the right.
- The horizontal axis represents the total labor supply L .

Prices, Wages, and Labor Allocation (4 of 10)

- The two sectors must pay the same wage because labor can move between sectors.
- If the wage were higher in the cloth sector, workers would move from making food to making cloth until the wages become equal.
 - Or if the wage were higher in the food sector, workers would move in the other direction.
- Where the labor demand curves intersect gives the equilibrium wage and allocation of labor between the two sectors.

Figure 4.4 The Allocation of Labor



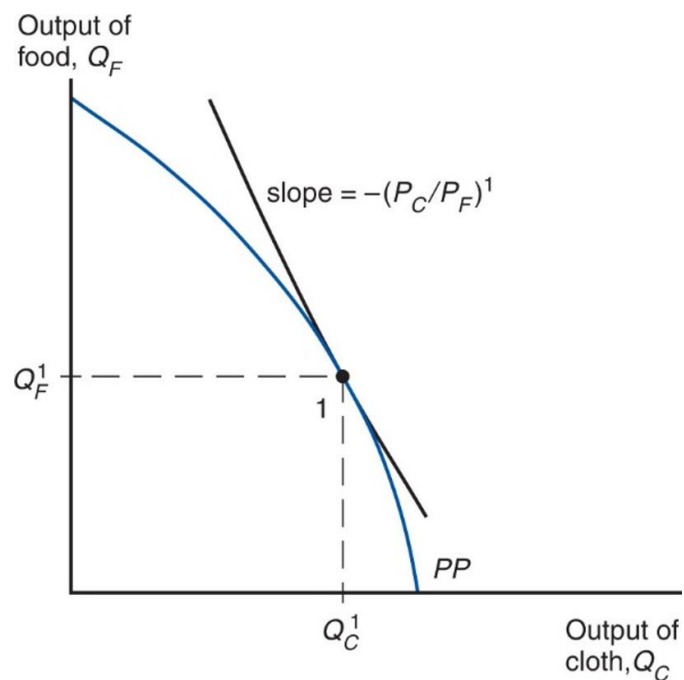
Labor is allocated so that the value of its marginal product ($P \times MPL$) is the same in the cloth and food sectors. In equilibrium, the wage rate is equal to the value of labor's marginal product.

Prices, Wages, and Labor Allocation (5 of 10)

- At the production point, the production possibility frontier must be tangent to a line whose slope is minus the price of cloth divided by that of food.
- Relationship between relative prices and output:

$$-\frac{MPL_F}{MPL_C} = -\frac{P_C}{P_F}$$

Figure 4.5 Production in the Specific Factors Model



The economy produces at the point on its production possibility frontier (PP) where the slope of that frontier equals minus the relative price of cloth.

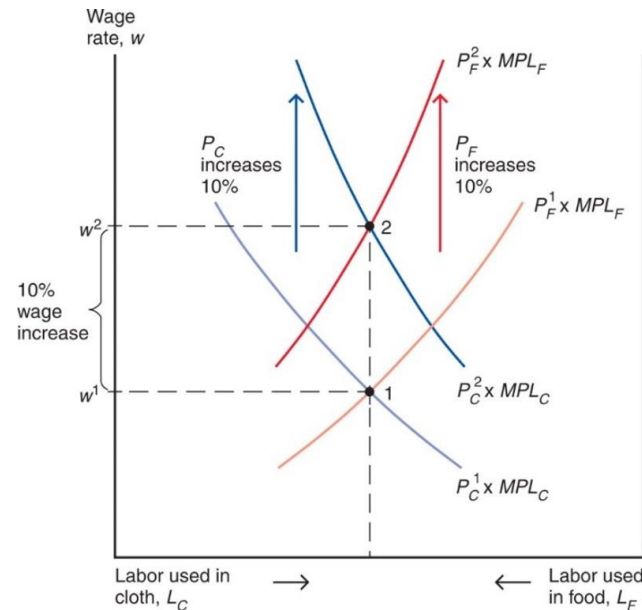
Prices, Wages, and Labor Allocation (6 of 10)

- What happens to the allocation of labor and the distribution of income when the prices of food and cloth change?
- Two cases:
 1. An equal proportional change in prices
 2. A change in relative prices

Prices, Wages, and Labor Allocation (7 of 10)

- When both prices change in the same proportion, no real changes occur.
 - The wage rate (w) rises in the same proportion as the prices, so real wages (i.e., the ratios of the wage rate to the prices of goods) are unaffected.
 - The real incomes of capital owners and landowners also remain the same.

Figure 4.6 An Equal-Proportional Increase in the Prices of Cloth and Food

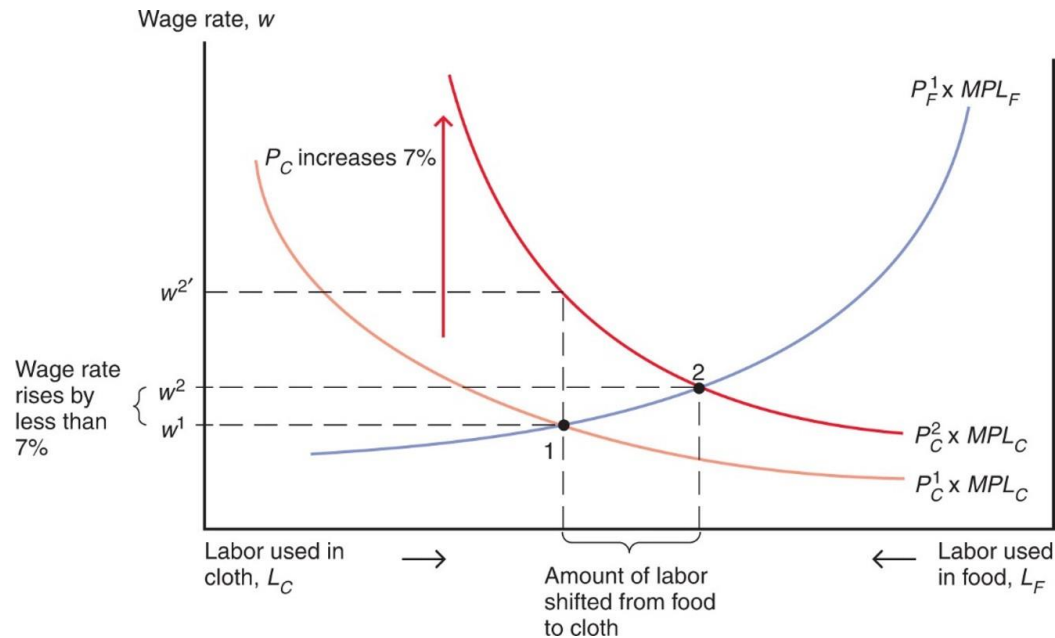


The labor demand curves in cloth and food both shift up in proportion to the rise in P_C from P_C^1 to P_C^2 and the rise in P_F from P_F^1 to P_F^2 . The wage rate rises in the same proportion, from w^1 to w^2 but the allocation of labor between the two sectors does not change.

Prices, Wages, and Labor Allocation (8 of 10)

- When only P_C rises, labor shifts from the food sector to the cloth sector and the output of cloth rises while that of food falls.
- The wage rate (w) does not rise as much as P_C since cloth employment increases and thus the marginal product of labor in that sector falls.

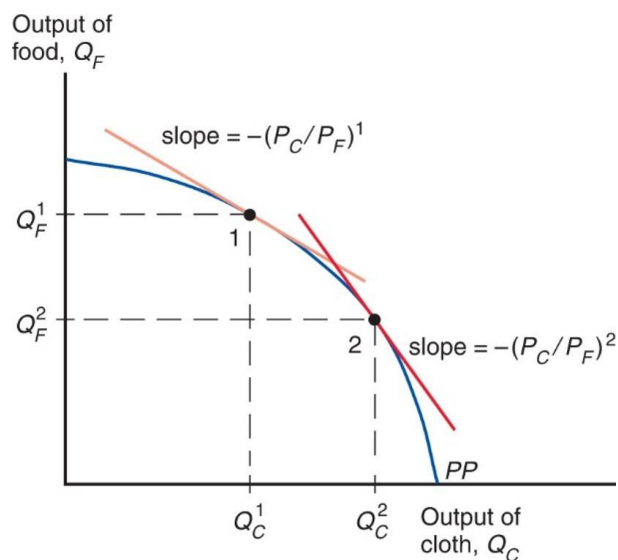
Figure 4.7 A Rise in the Price of Cloth



The cloth labor demand curve rises in proportion to the 7 percent increase in P_C , but the wage rate rises less than proportionately.

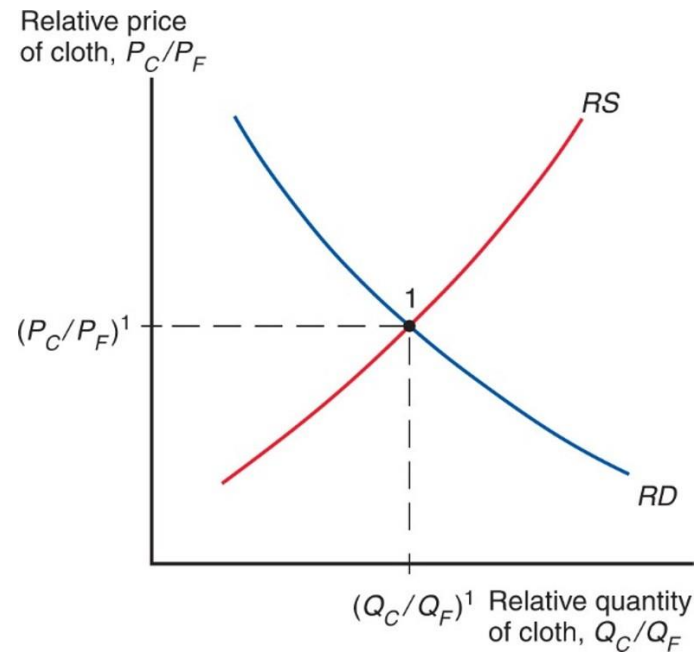
Labor moves from the food sector to the cloth sector. Output of cloth rises; output of food falls.

Figure 4.8 The Response of Output to a Change in the Relative Price of Cloth



The economy always produces at the point on its production possibility frontier (PP) where the slope of PP equals minus the relative price of cloth. Thus, an increase in $P_C > P_F$ causes production to move down and to the right along the production possibility frontier corresponding to higher output of cloth and lower output of food.

Figure 4.9 Determination of Relative Prices



In the specific factors model, a higher relative price of cloth will lead to an increase in the output of cloth relative to that of food. Thus, the relative supply curve RS is upward sloping. Equilibrium relative quantities and prices are determined by the intersection of RS with the relative demand curve RD .

Prices, Wages, and Labor Allocation (9 of 10)

- Relative Prices and the Distribution of Income
 - Suppose that P_C increases by 10%. Then, the wage would rise by less than 10%.
- What is the economic effect of this price increase on the incomes of the following three groups?
 - Workers, owners of capital, and owners of land

Prices, Wages, and Labor Allocation (10 of 10)

- Owners of capital are definitely better off.
- Landowners are definitely worse off.
- Workers: cannot say whether workers are better or worse off:
 - Depends on the relative importance of cloth and food in workers' consumption.